

MODULE 11 · DOMAIN 4 - RESPONSIBLE AI

Responsible AI

Domain 4 entire — and the vocabulary AWS expects you to use.

DOMAIN 4 of 5 EXAM WEIGHT 14% MODULE 11 of 14

Responsible AI is not a review at the end. It is a set of **decisions made throughout** — and each has a service attached.

AWS's eight dimensions of responsible AI

DIMENSION	WHAT IT MEANS
Fairness	Considering impacts on different groups of stakeholders.
Explainability	Understanding and evaluating system outputs.
Privacy and security	Appropriately obtaining, using and protecting data and models.
Safety	Preventing harmful system output and misuse.
Controllability	Mechanisms to monitor and steer system behaviour.
Veracity and robustness	Correct outputs even with unexpected or adversarial inputs.
Governance	Best practices across the AI supply chain.
Transparency	Enabling stakeholders to make informed choices.

ON THE EXAM

This is AWS's own published list, so questions use these exact words. Learn the eight names and one sentence each.

Where bias actually comes from

- ◆ **Training data.** Historical data records historical decisions, including unfair ones.
- ◆ **Representation.** A group with few examples gets worse predictions.
- ◆ **Labelling.** Human labellers bring their own assumptions.
- ◆ **Feedback loops.** The model's outputs shape the next dataset, compounding the bias.
- ◆ Removing the sensitive attribute does **not** remove bias. Proxies remain.

ON THE EXAM

That last point is tested. Deleting 'gender' from a dataset leaves job history, education and postcode, which still encode it.

Measuring and reducing bias

- ◆ **SageMaker Clarify** measures bias before training on the data, and after training on predictions.
- ◆ **Balanced, representative data** is the primary fix. Collect more of the underrepresented group.
- ◆ Resampling or reweighting can compensate when more data is unavailable.
- ◆ **Model Monitor** catches bias that appears after deployment.
- ◆ There is no single fairness definition. Different definitions genuinely conflict.

ON THE EXAM

'Ensure the training dataset represents all groups' is very often the correct answer. Data comes before algorithms.

Model Cards or AI Service Cards?

SAGEMAKER MODEL CARDS

Documents your model

- You create it for a model you built
- Intended use, training data, metrics
- Limitations and a risk rating
- Lives in your account, for your governance

AWS AI SERVICE CARDS

Documents Amazon's service

- Published by AWS, not by you
- Covers services like Rekognition and Titan
- Intended use cases, fairness, design choices
- Public documents you read, not write

THE TIEBREAKER

Yours or Amazon's? That single question separates them, and the exam asks it directly.

Bedrock Guardrails: six controls

- ◆ **Content filters** — hate, insults, sexual, violence, misconduct, at configurable strength.
- ◆ **Denied topics** — subjects the model must refuse, defined in natural language.
- ◆ **Word filters** — profanity plus your own blocked terms, such as competitor names.
- ◆ **Sensitive information filters** — detect PII and block or mask it.
- ◆ **Contextual grounding** — checks the answer is supported by the source and relevant to the question.
- ◆ **Prompt attack filters** — detect injection and jailbreak attempts.

ON THE EXAM

Guardrails apply to input and output, and work across any Bedrock model, agents and knowledge bases.

Contextual grounding deserves its own note

- ◆ Scores each response for **grounding** — is it supported by the retrieved source?
- ◆ And for **relevance** — does it actually address the question asked?
- ◆ Responses below your threshold are blocked rather than returned.
- ◆ This is a hallucination control that operates at runtime, not at training time.

ON THE EXAM

RAG reduces hallucination by supplying facts. Contextual grounding catches the cases where the model ignored them anyway.

Keeping a human in the loop

- ◆ **Amazon A2I** routes low-confidence predictions to human reviewers.
- ◆ **Return of control** stops an agent acting until your application approves.
- ◆ Human review is required wherever a wrong decision is not recoverable.
- ◆ Reviewer decisions become training data, improving the model over time.
- ◆ Full automation is a choice, not a default. The exam expects you to know when to refuse it.

ON THE EXAM

Medical, legal, financial and employment decisions are the classic 'human must review' scenarios.

Protecting people inside the data

- ◆ **Amazon Comprehend** detects and redacts PII in text before it reaches a model.
- ◆ **Amazon Macie** discovers sensitive data sitting in S3.
- ◆ **Guardrails** can mask PII in model output as a second line of defence.
- ◆ Data minimisation: do not collect what the use case does not need.
- ◆ Training on personal data without a lawful basis is a governance failure, not a technical one.

ON THE EXAM

PII appears at three points: in stored data, in prompts, and in outputs. Different services cover each.

The environmental dimension

- ◆ Training large models consumes substantial energy. Inference at scale consumes more over time.
- ◆ **Use the smallest model that does the job.** The single biggest lever.
- ◆ Prefer an existing pre-trained model over training your own.
- ◆ Batch inference and right-sized instances reduce waste.
- ◆ Managed and serverless services improve utilisation compared to idle capacity.

ON THE EXAM

Asked how to reduce the environmental impact of a generative AI workload, the answer is almost always the smaller model.

Turning principles into process

- ◆ Define acceptable use, approval gates and accountable owners before building.
- ◆ **Model Registry** holds approval status; nothing reaches production unsigned.
- ◆ **SageMaker Catalog** governs who may use which dataset or model.
- ◆ **AWS Audit Manager, Artifact and CloudTrail** supply the evidence trail.
- ◆ Governance covers the whole supply chain, including models you did not build.

ON THE EXAM

Governance questions ask about process and accountability, not about a single service. Look for the option describing a policy and an owner.

Carry these into the exam

- ◆ Eight dimensions: fairness, explainability, privacy and security, safety, controllability, veracity and robustness, governance, transparency.
- ◆ Removing a sensitive attribute does not remove bias. Proxies survive.
- ◆ Clarify measures bias and explains predictions. Model Monitor catches drift and post-deployment bias.
- ◆ Model Card is yours. AI Service Card is Amazon's.
- ◆ Guardrails: content filters, denied topics, word filters, PII, contextual grounding, prompt attacks.
- ◆ Reducing environmental impact usually means the smaller model.

A company's CV screening model advances men at a noticeably higher rate than women. The team removes the gender field from the training data and retrains. The disparity barely changes.

Why, and what should they do?

- ☐ A The model is broken; rebuild it from scratch
- ☐ B Proxy variables still encode gender; analyse with SageMaker Clarify
- ☐ C Increase the training epochs
- ☐ D Lower the decision threshold for women

◆ PAUSE AND DECIDE

Why, and what should they do?

- A

 Rebuild from scratch
Same data, same bias. Nothing would change.
- B

 Proxies; use Clarify
Career gaps, schools and hobbies still encode gender.
- C

 More epochs
Trains harder on biased data. Makes it worse.
- D

 Different thresholds by gender
Directly discriminatory, and likely unlawful.

ON THE EXAM

Deleting the column is the intuitive fix and it does not work. That is exactly why the exam asks it.

A retail chatbot built on Bedrock must never discuss competitor products, never give medical advice, and never repeat customer card numbers back in a reply.

The team wants one mechanism covering all three.

What should they configure?

- ☐ A A longer system prompt listing the rules
- ☐ B Bedrock Guardrails with denied topics, word filters and PII filters
- ☐ C Fine-tune the model to refuse those topics
- ☐ D Amazon Macie

◆ PAUSE AND DECIDE

What should they configure?

- A

 Longer system prompt
Helps, but is defeated by injection and not enforced.
- B

 Guardrails
Denied topics, word filters and PII filters cover all three.
- C

 Fine-tune to refuse
Expensive, slow, and still not a guarantee.
- D

 Amazon Macie
Finds sensitive data in S3. Not a chatbot control.

ON THE EXAM

Three requirements map to three Guardrails features. When a scenario lists several rules, one policy usually covers them.

An insurer built its own claims triage model on SageMaker. An internal audit committee wants a document describing the model's intended use, its training data, its performance and its known limitations.

They ask which AWS artefact provides this.

What should the team produce?

- ☐ A An AWS AI Service Card
- ☐ B A SageMaker Model Card
- ☐ C A CloudTrail export
- ☐ D An AWS Artifact report

◆ PAUSE AND DECIDE

What should the team produce?

- A

 AI Service Card
Published by AWS about AWS services. Not your model.
- B

 SageMaker Model Card
Documents a model you built: use, data, metrics, limits.
- C

 CloudTrail export
API call logs. Not model documentation.
- D

 AWS Artifact report
AWS compliance reports, such as SOC and ISO.

ON THE EXAM

The insurer built the model, so the document is theirs to write. AI Service Cards are Amazon's, about Amazon's services.

A pharmaceutical company runs a RAG assistant over its approved drug information. Occasionally the model produces confident answers that go beyond what the retrieved documents actually say.

Any statement not supported by the source is a regulatory problem.

Which control addresses this specifically?

- | | |
|--|---|
| ☐ A Guardrails contextual grounding check | ☐ B Guardrails denied topics |
| ☐ C Lower the temperature | ☐ D SageMaker Model Monitor |

◆ PAUSE AND DECIDE

Which control addresses this specifically?

- A

 Contextual grounding
Scores whether the answer is supported by the source, and blocks it.
- B

 Denied topics
Blocks subjects. The subject here is correct.
- C

 Lower temperature
Reduces variation. No guarantee of grounding.
- D

 Model Monitor
Watches SageMaker models for drift. Not per-response checking.

ON THE EXAM

RAG already supplies the facts. The failure is the model going beyond them, which is exactly what grounding checks measure.

A sustainability team is asked to reduce the environmental footprint of a generative AI feature that summarises support tickets overnight.

Quality requirements are modest and there is no latency constraint.

Which change helps most?

- ☐ A Move to a larger, more capable model
- ☐ B Use a smaller model and run it as a batch job
- ☐ C Increase the context window
- ☐ D Fine-tune the existing model

◆ PAUSE AND DECIDE

Which change helps most?

- A

 Larger model
More compute per request. The wrong direction.
- B

 Smaller model, batch
Less compute per token, better utilisation, no idle capacity.
- C

 Larger context window
More tokens processed per call, so more energy.
- D

 Fine-tune
Adds a training cost and standing hosted capacity.

ON THE EXAM

Modest quality needs plus no latency constraint is the exam telling you a smaller model and batch are both available.

A hospital deploys a model that reads scans and proposes a preliminary finding. The finding is written into the patient record and visible to clinicians.

The clinical governance board asks what safeguard is required before go-live.

What is the correct answer?

- ☐ A Increase model accuracy above 99%
- ☐ B A human review step before any finding is recorded
- ☐ C Enable Guardrails content filters
- ☐ D Publish an AI Service Card

◆ PAUSE AND DECIDE

What is the correct answer?

- A

 Accuracy above 99%
No accuracy figure makes an unreviewed clinical finding acceptable.
- B

 Human review before recording
Controllability. A clinician decides, the model assists.
- C

 Content filters
Blocks harmful content. Clinical findings are not harmful content.
- D

 AI Service Card
Documentation of an AWS service. Not a safeguard.

ON THE EXAM

Option A is the trap for people who think safety is a threshold. In high-stakes decisions, oversight is structural, not statistical.

A bank wants to analyse customer complaint emails for themes. The emails contain names, account numbers and addresses.

Policy states personal data must be removed before any text reaches a foundation model.

Which approach fits the policy?

- ☐ A Amazon Macie on the S3 bucket
- ☐ B Amazon Comprehend PII detection and redaction before the model call
- ☐ C Guardrails PII filter on the model output
- ☐ D Encrypt the emails with KMS

◆ PAUSE AND DECIDE

Which approach fits the policy?

- A

 Macie on the bucket
Discovers sensitive data. Does not redact before a call.
- B

 Comprehend redaction first
Removes PII before the text reaches the model.
- C

 Guardrails on output
Too late. The PII already reached the model.
- D

 Encrypt with KMS
Protects at rest. Decrypted text still reaches the model.

ON THE EXAM

The word 'before' decides it. Option C is a genuine control in the wrong position, which is the subtlest kind of wrong answer.

A company is documenting how its AI system behaves when it receives deliberately misleading or malformed input, and how consistently it produces correct output under those conditions.

It wants to record this against AWS's responsible AI framework.

Which dimension does this describe?

- | | |
|--|--|
| ☐ A Transparency | ☐ B Controllability |
| ☐ C Veracity and robustness | ☐ D Governance |

◆ PAUSE AND DECIDE

Which dimension does this describe?

- A

 Transparency
Helping stakeholders make informed choices.
- B

 Controllability
Mechanisms to monitor and steer behaviour.
- C

 Veracity and robustness
Correct output even with unexpected or adversarial input.
- D

 Governance
Best practices across the AI supply chain.

ON THE EXAM

AWS's definition is 'correct system outputs, even with unexpected or adversarial inputs'.
The scenario is that sentence rephrased.